

IDEA: A Real-Time Unified Accelerator for Dual-Task Image Restoration: Dehazing and Illumination Enhancement

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Abstract—High-performance image dehazing is crucial for various visual tasks, particularly in the field of embodied intelligence, where real-time processing and energy efficiency on edge devices are essential. This paper introduces a deep learning-based dehazing network, namely DIE, and its corresponding accelerator, IDEA, which are optimized to correct illumination bias. The proposed DIE features a Chunk-Wise Fused Concatenation Scheduling (CFCS) and an inter-Processing Element line buffer (ILB) mechanism to enhance data reuse, leading to a 2.24× acceleration. Additionally, a dual-task scheduling approach and a Heterogeneous Ping-Pong (HTTP) strategy are utilized to balance the workload and improve parallelism. Implemented in a 55 nm process, the accelerator achieves a core area of 2.83 mm², power consumption of 192 mW, and a frame rate of 72 fps at 1080p resolution. To the best of our knowledge, this is the first ASIC implementation of a deep learning-based dehazing architecture, demonstrating strong potential for low-power, high-quality visual enhancement in edge intelligence applications.

Index Terms—Dehazing Accelerator, Illumination Correction, Layer Fusion, Efficient Mapping

I. INTRODUCTION

Image restoration, as a crucial preprocessing step in visual perception systems, not only improves image visibility but also enhances the performance of downstream tasks such as object detection and scene understanding. For example, the image dehazing has been increasingly adopted in autonomous driving, UAV-based imaging, and embodied intelligence.

Deep neural network (DNN)-based dehazing methods have demonstrated superior restoration quality compared to traditional priors-based algorithms [1]–[3]. High-parameter models such as FFA-Net [4] and DeHamer [5] achieve remarkable haze removal performance by leveraging transformer-based or hybrid CNN–ViT architectures. However, their computational complexity and large parameter size make them impractical for real-time deployment on edge hardware. Consequently, most hardware-oriented designs adopt DCP-based or lightweight DNN-based dehazing methods.

Among lightweight solutions, concat-based architectures, such as AOD-Net [6], estimate transmission maps using multiple receptive fields and atmospheric scattering modeling. This structure often introduces inconsistencies in brightness and color due to limited feature representation. Hardware designers have noticed this issue and attempted to compensate for it. For instance, Tan et al. [7] utilized mixed atmospheric priors from multi-frame videos to recover foreground illumination,

while Zhang et al. [8] employed 3D-LUT conversion from RGB to HSI space for color correction. However, these efforts treat illumination enhancement as an isolated post-processing step, overlooking the intrinsic correlation between dehazing and illumination recovery. Recent findings, such as those by Galdran et al. [9], further reveal that illumination enhancement and dehazing share common physical principles in reflection modeling, indicating the potential of a joint dehazing–illumination framework for both visual quality and hardware efficiency.

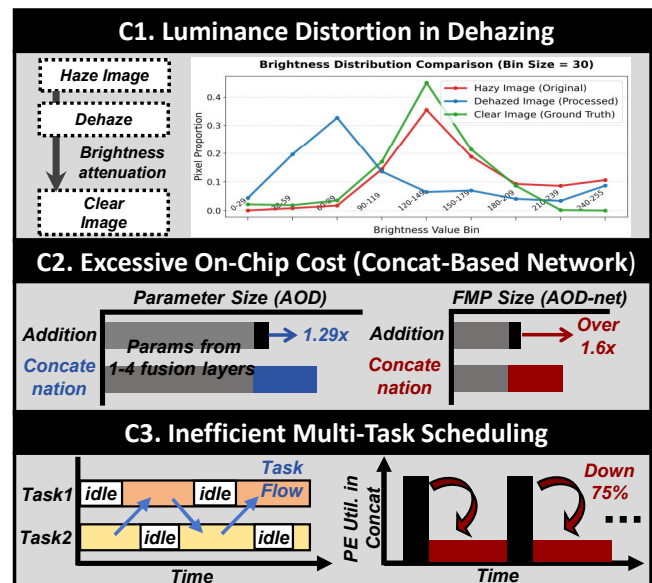


Fig. 1. Challenges of Lightweight Dehazing Neural Network Acceleration.

As shown in Fig. 1, there are three challenges in developing a lightweight dehazing and illumination enhancement system for downstream vision tasks like embodied intelligence. 1) The dehazing process inherently alters the original luminance distribution of images and tends to redistribute luminance peaks, often resulting in brightness distortion or color imbalance in the restored images. 2) Concat-based architectures require significantly larger on-chip SRAM and computation resources compared with add-based networks. Although concat-based networks have demonstrated superior global context modeling ability in lightweight dehazing tasks [6], the increased number of parameters and the need to store multiple intermediate fea-

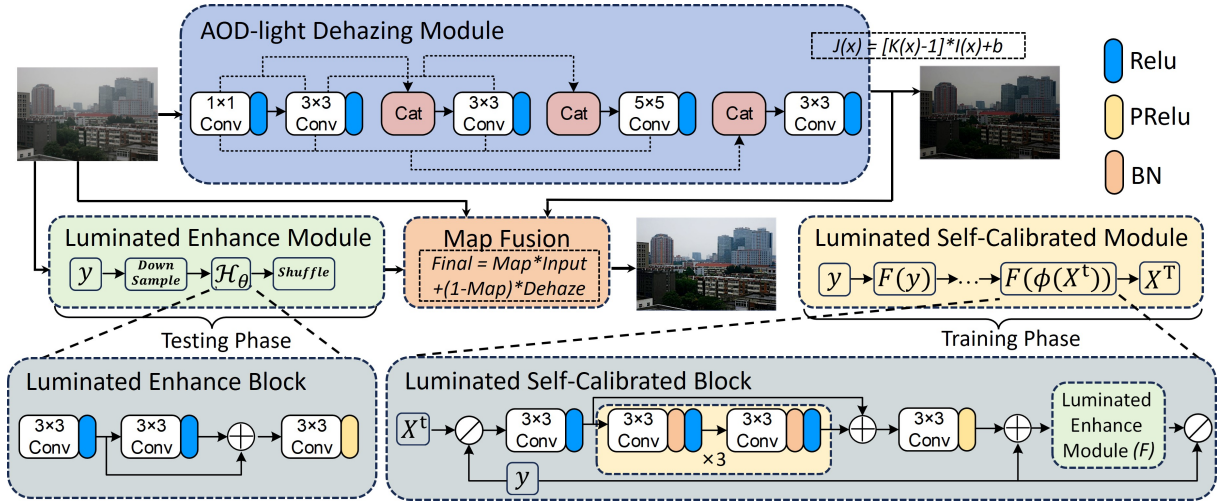


Fig. 2. Dehazing and Illumination Enhancement (DIE) Network.

ture maps contradict the lightweight design objective, leading to slower inference and higher memory demand. 3) Multi-task scheduling mechanism suffers from low PE utilization and inefficient on-chip memory reuse. Previous studies [10] have proposed adaptive task allocation. However, sequential dependencies lead to idle resources in serially coupled tasks.

In this work, 1) we propose the IDEA architecture and DIE Network, mitigating color distortion while maintaining lightweight computation through a coupled dual-module design that achieves end-to-end high-fidelity restoration. 2) To accelerate a concat-based network, a Depth-Time-Chunk (DTC) strategy is introduced, in which the chunk-wise fusion mechanism improves computation speedup by 2.24x, and the inter-PE line-buffer optimization reduces computation cost and on-chip memory. 3) A DIE-oriented dual task optimization scheme is proposed, featuring a task-aware PE-mapping strategy and a heterogeneous buffer system. These innovations enable efficient real-time enhancement for downstream vision tasks under adverse visibility and illumination conditions.

II. HARDWARE ORIENTED ALGORITHM DESIGN

A. DLE Algorithm Structure and Flow

To enhance image restoration quality while maintaining a lightweight design, we propose the Dehazing-Illuminated-Enhancement (DIE) network, as illustrated in Fig. 2. The first stage, named the AOD-light dehazing module, inherits the concatenation-based feature aggregation from AOD-Net [6] but introduces a hardware-coordinated convolution design to improve parameter efficiency. Specifically, multiple convolutional layers with different kernel sizes are retained to ensure diverse receptive fields, while kernel dimensions are carefully optimized to account for hardware constraints. Compared with the original network, the proposed design achieves a 43% reduction in parameter count, with less than 0.2 dB PSNR degradation on the NYU2 dataset.

The dehazed results are subsequently processed by the Luminance Enhancement Module, which compensates for brightness attenuation introduced during atmospheric scattering correction. This module is inspired by Self-Calibrated Illumination

(SCI) [11] but is redesigned for lightweight hardware deployment. During training, an auxiliary illumination calibration branch is introduced that shares weights with the main path to improve convergence speed and robustness. Furthermore, a two-stage resolution adjustment is employed: the input image is first downsampled by a factor of two, then expanded in the channel dimension from C to 4C, and finally reconstructed using a PixelShuffle upsampling unit. This structure achieves a 2.1 dB improvement in PSNR on the DarkFace dataset under low-light conditions. The outputs of two modules, together with the input image, are fed into the fusion module. The adoption of the dehazing result is determined by the reflection component r [11].

B. Hardware Implementation

During hardware deployment, each input image is partitioned into multiple small patches, which are sequentially loaded into on-chip memory for convolutional processing. Each patch is 30×30 and includes boundary strips required for convolutional continuity. Both feature maps and weights are quantized to 16-bit precision. For a 1080P image, this patch-based inference strategy reduces on-chip memory requirements by approximately 99.5%.

III. HARDWARE ARCHITECTURE

A. Overview

The proposed DIE algorithm is deployed on the IDEA accelerator. The architecture is illustrated in Fig 3. The architecture

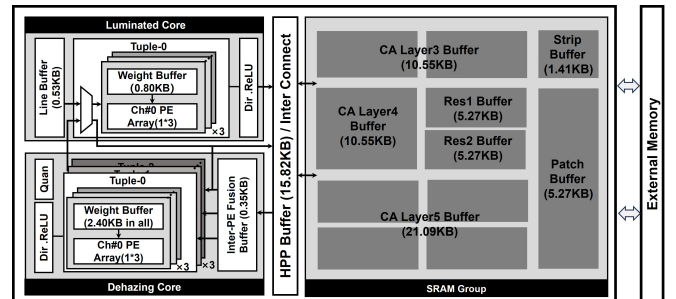


Fig. 3. Overview of the IDEA Architecture.

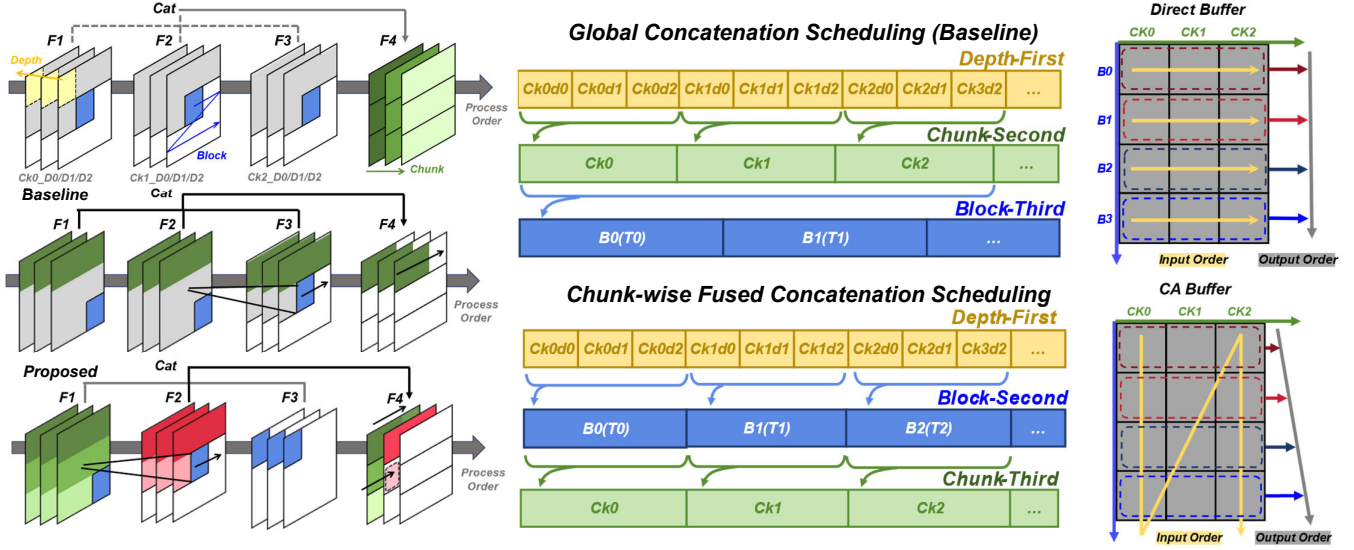


Fig. 4. Chunk-Wise Fused Concatenation Scheduling (CFCS) and CA Buffer Design.

consists of two task-specific cores, the Dehazing Core and the Luminance Core, each with a different number of PE arrays. Redundant resources among cores are dynamically scheduled to accelerate other tasks when available. Partial sums (Psums) are computed at full precision and quantized to 16-bit on-chip. Feature maps (FMPs) are preloaded into the Line Buffer, and weights for each PE are stored in the Weight Buffer, both following a weight-stationary dataflow. The Line Fusion Buffer supports on-chip layer fusion as well as conventional line buffer functions.

For memory, the Patch/Strip Buffer stores input image patches, while the Concat Layer Buffer (CA Buf) and Residual Buffer (Res Buf) temporarily hold intermediate results according to the computational strategy. Selected outputs are routed to the Line Buffer through the Heterogeneous Ping-Pong Buffer (HPPB) to facilitate efficient data reuse.

B. Chunk-wise Fused Concatenation Scheduling

To address inefficiency in conventional global concatenation, we propose a Chunk-wise Layer Fusion Scheduling method, as illustrated in Fig. 4. In the baseline global scheduling, computations follow a depth-first, chunk-second, block-third order, where all concatenation operations are deferred to the end. This accumulation leads to severe latency and unbalanced workloads across PE arrays.

The proposed strategy partitions the final concatenation into several chunk-level fusion subtasks (depth-first, block-second, chunk-third order). Each chunk performs local fusion immediately after completing its preceding FMP computation, and partial results are temporarily stored in the CA Buffer, which also defines the data sequence for the Line Buffer to maintain continuous streaming without pipeline stalls. This chunk-wise fusion converts a large single-stage concatenation into multiple low-latency fusion steps. Without extra on-chip memory cost, the CA Buffer reuses existing storage to perform both fusion and temporary caching. For example, in the Dehazing Module, redundant concatenation layers (Layers 3–5) are skipped, reducing the computation time of Layers 3–4

by 50% and of Layer 5 by 75%, achieving an overall 55.6% reduction in the Dehazing Core execution time.

C. Layer Fusion Optimization with Inter-PE Line buffer

During chunk-wise fusion, multiple two-layer 3×3 convolution fusions are repeatedly executed. Figure 5 compares the baseline Naive Triple-PE Fusion (NTF) with the proposed Inter-PE Line Buffer (ILBF) scheme. In NTF, three 3×3 convolution groups are cascaded, where Layer1 outputs are directly forwarded to Layer2, and Layer3 accumulates the results. Although this design eliminates off-chip access, frequent inter-layer updates still cause redundant computations.

The proposed ILBF introduces an inter-PE line buffer to reuse Psums across layers. Two previously computed rows are buffered, while a single PE group processes the new row per cycle. These three rows form the convolution window for the next layer, enabling efficient local reuse without pipeline stalls. Compared with NTF, ILBF reduces fusion-stage computation by 50% while requiring only 0.117KB of additional on-chip memory, achieving high efficiency with negligible memory overhead.

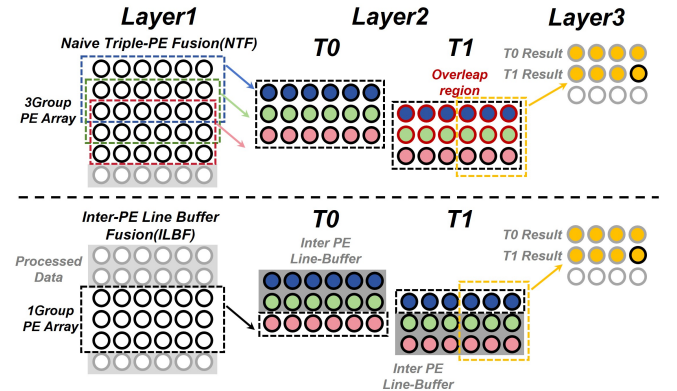


Fig. 5. Inter-PE Line Buffer Fusion (ILBF).

D. Multi-task Schedule

To address the computational imbalance stemming from the sequential dependency between the dehazing and enhancement

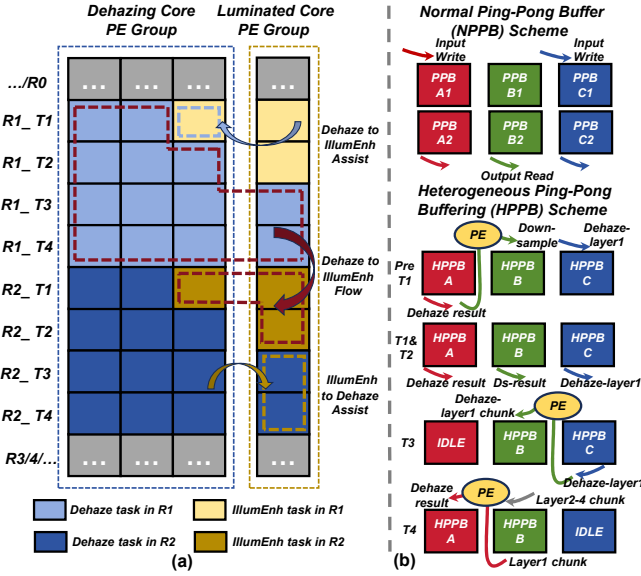


Fig. 6. (a) Dual-Task Scheduling and (b) Heterogeneous PPB Scheme.

tasks—where the former is backend-heavy and the latter is frontend-heavy—we propose the IDEA-PE scheduling strategy, illustrated in Fig. 6(a). This strategy, enabled by inter-core connections, features two synergistic modes: Assist and Flow. The Assist mode facilitates dynamic, cross-core resource allocation within each cycle R. Specifically, during the T1 phase, a set of PEs from the Dehazing core is reallocated to support the fused input and enhancement convolution of the IllumEnh task. Conversely, during the T3 and T4 phases, PEs from the Luminated core are assigned to compute the 5x5 convolution for the Dehaze task, with three PEs collaborating on each convolution. The Flow mode then directs the intermediate results from the Dehaze core to the Luminated core, orchestrated by the HPPB policy. This fine-grained mapping strategy boosts PE utilization to 100% across all stages.

Furthermore, the HPPB policy, detailed in Fig. 6(b), not only enables efficient dataflow but also unifies the management of three distinct image inputs (for dehazing, atmospheric light, and illumination downsampling), achieving 92% memory utilization. Compared to a baseline design employing three separate ping-pong buffers, our HPPB policy reduces the corresponding on-chip memory footprint by 50%.

IV. EXPERIMENT AND EVALUATION

The IDEA architecture is designed in Verilog and synthesized for the UMC 55nm CMOS process. The proposed DIE network is trained on the patched NYU2 and MIT datasets. Table I presents a comparison between the proposed DIE algorithm and widely adopted dehazing networks on the NYU2 and SOTS datasets. The DIE network demonstrates superior performance in terms of both PSNR and SSIM. Specifically, compared with the indoor NYU2 dataset, DIE achieves a 1.3dB increase in PSNR, and a larger 2.8dB improvement is observed on the SOTS outdoor dataset. This performance enhancement is attributed to the illumination map mapping strategy, which effectively avoids misclassifying highlight ar-

TABLE I
COMPARISON OF THE DEHAZE ALGORITHMS

Method	Parameter	MAC (GOPS)	NYU2		SOTS-O	
			PSNR	SSIM	PSNR	SSIM
DCP [3]	-	-	19.07	0.8235	18.38	0.8179
MSCNN [12]	8.23k	25.08	19.11	0.8295	17.57	0.8102
DehazeNet [13]	8.56k	37.05	18.85	0.7749	22.26	0.8516
AOD-Net [6]	1.76k	7.26	19.69	0.8478	20.08	0.8504
DIE(Ours)	1.64k	5.13	20.99	0.8589	22.88	0.8628

*Input image: 1080P. Blue: Best result. Green: Second best result.

TABLE II
DEHAZING ARCHITECTURE COMPARISON.

Works	17'TVLSI [14]	23'TCAD [7]	23'TVLSI [15]	IDEA (Proposed)
Device/Technology	Stratix	XC7Z100	XC7Z020	UMC-55nm
Alg. Method	DCP-based	Prior-based	DCP-based	DIE
PSNR ¹ [dB]	16.09	16.30 ²	20.05	22.36
Nor. Area (mm ²)	-	-	-	2.83
Frequency (MHz)	116	200	85.2	200
Power (mW)	-	5900	-	192
SRAM Size (KB)	-	188	-	79.32
Throughput (Mpixels/s)	116	80.8	85.2	149.5
Energy Efficiency ³ (mJ/frame)	-	-	-	2.67
Frame Rate (fps) ³	56	39	41	72

¹ PSNR is measured in SOTS dataset with 16-bit quantized

² MPAD's PSNR is measured in HAZE-O dataset

³ Normalized Input image to 1080P.

eas (e.g., outdoor sky) and low-fog/high-light contrast regions (e.g., building reflections) as haze, thus preventing potential peak clipping.

Table II compares the proposed IDEA with other dehazing architectures, demonstrating the highest throughput and the smallest on-chip SRAM size. Thanks to the chunk-wise layer fusion, IDEA performs computation optimization in both the group level (multiple FMPs) and row level (single FMP), enabling a 1.28–1.84 *times* performance improvement over conventional multi-stage pipelined VLSI dehazing designs. By integrating the neural network-based DIE algorithm, IDEA also surpasses all existing DCP-based architectures by more than 2dB in reconstruction quality. The proposed multi-task scheduling further improves energy efficiency, reaching 2.67 mJ/frame in pre-silicon evaluation. As far as we know, this is the first ASIC implementation of a neural network for image dehazing; therefore, an absolute energy-efficiency comparison is not yet available and will be discussed in future dehazing-oriented ASIC studies.

V. CONCLUSION

In this work, we proposed a hardware-friendly dehazing network (DIE) with illumination enhancement and implemented its ASIC architecture (IDEA), which features layer fusion, memory optimization, and task scheduling. The architecture achieves 72 fps at 1080p with a 2.83 mm² area and 192 mW power consumption, marking, to our knowledge, the first ASIC-based neural network for haze removal.

ACKNOWLEDGEMENT

This work was supported by the National Science Foundation of China under Grant No. 62574032. Liang Chang is the corresponding author. Email: liangchang@uestc.edu.cn.

REFERENCES

- [1] J.-H. Kim, W.-D. Jang, J.-Y. Sim, and C.-S. Kim, "Optimized contrast enhancement for real-time image and video dehazing," *Journal of Visual Communication and Image Representation*, vol. 24, no. 3, pp. 410–425, 2013.
- [2] Q. Zhu, J. Mai, and L. Shao, "A fast single image haze removal algorithm using color attenuation prior," *IEEE transactions on image processing*, vol. 24, no. 11, pp. 3522–3533, 2015.
- [3] K. He, J. Sun, and X. Tang, "Single image haze removal using dark channel prior," *IEEE transactions on pattern analysis and machine intelligence*, vol. 33, no. 12, pp. 2341–2353, 2010.
- [4] X. Qin, Z. Wang, Y. Bai, X. Xie, and H. Jia, "Ffa-net: Feature fusion attention network for single image dehazing," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 34, no. 07, 2020, pp. 11 908–11 915.
- [5] C.-L. Guo, Q. Yan, S. Anwar, R. Cong, W. Ren, and C. Li, "Image dehazing transformer with transmission-aware 3d position embedding," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 5812–5820.
- [6] B. Li, X. Peng, Z. Wang, J. Xu, and D. Feng, "Aod-net: All-in-one dehazing network," in *Proceedings of the IEEE international conference on computer vision*, 2017, pp. 4770–4778.
- [7] Y. Tan, Y. Zhu, Z. Huang, H. Tan, and K. Li, "Mapd: an fpga-based real-time video haze removal accelerator using mixed atmosphere prior," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 42, no. 12, pp. 4777–4790, 2023.
- [8] J. Zhang, D. Fan, and L. Chang, "Hiera: High-quality and high-throughput dehazing hardware accelerator with reconfigurable computing unit," in *2024 IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*. IEEE, 2024, pp. 75–80.
- [9] A. Galdran, A. Alvarez-Gila, A. Bria, J. Vazquez-Corral, and M. Bertalmio, "On the duality between retinex and image dehazing," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 8212–8221.
- [10] X. Zhao, Z. Hu, Z. Guo, H. Fan, X. Yang, J. Zhou, and L. Chang, "A rram-based high energy-efficient accelerator supporting multimodal tasks for virtual reality wearable devices," in *Proceedings of the 61st ACM/IEEE Design Automation Conference*, 2024, pp. 1–6.
- [11] L. Ma, T. Ma, R. Liu, X. Fan, and Z. Luo, "Toward fast, flexible, and robust low-light image enhancement," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 5637–5646.
- [12] W. Ren, S. Liu, H. Zhang, J. Pan, X. Cao, and M.-H. Yang, "Single image dehazing via multi-scale convolutional neural networks," in *European conference on computer vision*. Springer, 2016, pp. 154–169.
- [13] B. Cai, X. Xu, K. Jia, C. Qing, and D. Tao, "Dehazenet: An end-to-end system for single image haze removal," *IEEE transactions on image processing*, vol. 25, no. 11, pp. 5187–5198, 2016.
- [14] B. Zhang and J. Zhao, "Hardware implementation for real-time haze removal," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 25, no. 3, pp. 1188–1192, 2016.
- [15] B. B. Upadhyay and K. Sarawadekar, "Vlsi design of saturation-based image dehazing algorithm," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 31, no. 7, pp. 959–968, 2023.